

ME310 Group Project: Ultrasonic Sensor

04/29/26

Bryson Whitaker, Benjamin Durkin, Mathis Seidl, Luis Macias

Group 2

Objectives

The primary objective of this experiment is to utilize an ultrasonic air-ranging sensor to measure the displacement of an oscillating mass in a second-order spring-damper system. The experiment further aims to determine the system's natural frequency and damping ratio through multiple methods, and to characterize the frequency response of the oscillator across a range of driving frequencies spanning below and above resonance. Finally, the collected displacement data will be analyzed and compared against theoretical predictions from linear second-order oscillator theory to identify any deviations arising from system non-linearities.

Theory

Fundamental Principles of Sound and Ultrasound

Sound is a mechanical wave that propagates through a medium via the periodic compression and rarefaction of molecules. These pressure disturbances travel outward from a source at a speed determined by the physical properties of the medium, namely its elasticity and density. For an ideal gas, the speed of sound is given by:

$$c = \sqrt{\gamma \frac{P}{\rho}} \quad (1)$$

where γ is the ratio of specific heats (approximately 1.4 for air), P is the ambient pressure, and ρ is the air density. This can equivalently be expressed as a function of temperature:

$$c = \sqrt{\gamma R T} \quad (2)$$

where R is the specific gas constant for air and T is the absolute temperature in Kelvin. At standard room temperature (approximately 20°C), the speed of sound in air is 343 m/s, though this value increases with temperature at a rate of about 0.6 m/s per degree Celsius. This temperature dependence is an important practical consideration: if the ambient temperature in the laboratory fluctuates significantly during data acquisition, the effective speed of sound and therefore the distance calculation may drift slightly, introducing a small but measurable systematic error into the displacement measurements.

"Ultrasound" refers to acoustic waves at frequencies above the threshold of human hearing, generally taken to be above 20 kHz. The ultrasonic sensor used in this experiment operates at a center frequency at or near the transducer's mechanical resonance frequency, which is typically in the range of 40–200 kHz for air-coupled

sensors. The ultrasonic sensor used in the lab operates at around 200kHz. Operating at such high frequencies offers several practical advantages: the shorter wavelength improves spatial resolution, the beam can be more tightly focused, and the signal is less susceptible to audible ambient noise in the lab environment.

Pulse-Echo Ranging: Time-of-Flight Measurement

The core operating principle of the ultrasonic sensor is the pulse-echo or time-of-flight (TOF) method. The transducer is periodically switched between two modes: a transmit ("pulse") mode in which a short burst of ultrasonic energy is emitted, and a receive ("echo") mode in which the transducer listens for the returning reflection. When the emitted pulse strikes a surface a portion of the acoustic energy is reflected back toward the transducer. The sensor's pulse and echo can be visualized via an ADC as they correspond to measurable voltages. The pulse must travel to the target and back, the total path length is $2d$, where d is the distance to the target. The distance is therefore calculated as:

$$d = \frac{c\Delta t}{2} \quad (3)$$

This deceptively simple relationship is the foundation of the entire measurement. The precision of the distance measurement is fundamentally limited by how accurately the time interval Δt can be resolved, which depends on the electronics, the sharpness of the emitted pulse, and the signal-to-noise ratio of the received echo. In practice, well-designed ultrasonic ranging systems can achieve millimeter-level resolution, which is more than adequate for measuring the oscillation amplitudes expected in this experiment.

Transducer Construction and Piezoelectric Effect

The ultrasonic transducer itself is constructed using a piezoelectric element; a material that mechanically deforms when an electric field is applied across it and generates an electric charge when mechanically strained. Common piezoelectric materials include lead zirconate titanate (PZT) ceramics and polyvinylidene fluoride (PVDF) films. In the transmit mode, an oscillating voltage at the transducer's resonance frequency is applied across the piezoelectric element, causing it to vibrate and radiate acoustic energy into the surrounding air. In the receive mode, the returning pressure wave mechanically deforms the piezoelectric element, generating a small analog AC voltage signal that can then be amplified and processed.

The efficiency of energy transfer between the transducer and the air is characterized by the acoustic impedance mismatch. Acoustic impedance Z is defined as:

$$Z = \rho c \quad (4)$$

where ρ is the medium density and c is the speed of sound. The large difference in acoustic impedance between the piezoelectric ceramic and air means that only a small fraction of the energy is transmitted into the air; most is reflected back into the element. To partially mitigate this, air-coupled transducers often incorporate matching layers or specialized focusing cones that improve coupling efficiency and directivity.

Beam Geometry and Directivity

The ultrasonic transducer does not emit energy uniformly in all directions. Instead, it produces a beam with a characteristic directivity pattern governed by the geometry of the emitting surface and the wavelength of the sound. For a circular piston-like transducer of radius a , the half-angle of the main beam lobe is approximately:

$$\theta = \sin^{-1}(0.61 \frac{\lambda}{a}) \quad (5)$$

where $\lambda = c/f$ is the acoustic wavelength. At 40 kHz in air, the wavelength is approximately 8.6 mm, so a transducer with a face diameter of a few centimeters produces a fairly narrow, well-collimated beam. This focused beam geometry is essential for this experiment: it ensures that the emitted pulse reflects cleanly off the oscillating mass and does not scatter off surrounding structures such as the shaft, the spring, or the lab bench, all of which could introduce spurious echoes and corrupt the distance measurement.

In addition to the main lobe, the transducer's directivity pattern includes side lobes at larger angles. While these carry much less energy than the main lobe, they can still produce weak reflections from off-axis objects. Careful alignment of the sensor directly above (or below) the oscillating mass, with the beam axis perpendicular to the reflecting surface, is therefore critical for obtaining clean, reliable measurements.

Reflection Characteristics and Target Surface Considerations

The strength of the echo received by the transducer depends heavily on the geometry and material properties of the reflecting surface. A flat, smooth, rigid surface oriented perpendicular to the beam axis produces a strong specular reflection in which most of the incident energy is reflected directly back toward the transducer. Conversely, a curved, rough, or obliquely oriented surface scatters the reflected energy in multiple directions, reducing the echo amplitude and potentially causing the sensor to miss detections. In this experiment, the reflecting target is the top collar of the oscillating mass. Since the collar is a machined metal surface, it should provide a relatively strong and consistent reflection, provided the sensor is well-aligned. However, as the mass oscillates vertically,

the reflecting surface remains nominally perpendicular to the beam axis, so this alignment condition should be maintained throughout the measurement.

Air absorption is another factor that can attenuate the echo signal, particularly at higher ultrasonic frequencies. The absorption coefficient of air increases with frequency, meaning that higher-frequency transducers suffer greater signal attenuation over a given distance. For the short distances involved in this experiment (likely on the order of a few centimeters to perhaps half a meter), air absorption is not expected to be a significant source of error, but it is worth noting as a general limitation of ultrasonic ranging at higher frequencies.

No Calibration Required and Advantages Over Other Sensors

One notable practical advantage of the ultrasonic sensor, as noted in the lab documentation, is that it does not require calibration in the same sense as, for example, the optical proximity sensor or the linear potentiometer. The distance measurement is derived directly from a physical constant and a time measurement, rather than from an empirical voltage-to-displacement relationship that must be established separately for each setup. This makes the ultrasonic sensor inherently more robust to sensor-to-sensor variability and simplifies the experimental workflow considerably.

Compared to a linear potentiometer, which requires physical contact (via an extensible wire) with the moving mass, the ultrasonic sensor is entirely non-contact. This is a significant advantage in a dynamic measurement: the sensor imposes no mechanical loading on the oscillating system, so it does not alter the mass, damping, or stiffness of the system being measured. The potentiometer's extensible wire, by contrast, adds a small but nonzero force to the mass and may introduce additional damping or non-linear behavior, particularly at higher oscillation amplitudes or frequencies.

Compared to the accelerometer, which measures acceleration and requires double integration to recover displacement the ultrasonic sensor measures displacement directly. This makes the subsequent data analysis more straightforward, as no integration is required and the accumulated error associated with integration is avoided. However, the ultrasonic sensor does carry its own additional analysis burden, as recognized by the additional project credit awarded for its use: the experimenter must carefully interpret the pulse timing, account for the speed of sound, and be vigilant about aliasing and spurious reflections.

Theory of a Second-Order System

A second-order mechanical system, such as the mass–spring–damper oscillator used in this experiment, is governed by the balance of inertial, damping, and restoring forces. Applying Newton’s Second Law to a mass m attached to a spring of stiffness k and a damper with coefficient c , the equation of motion is:

$$m\ddot{y} + c\dot{y} + ky = F(t) \quad (6)$$

where $y(t)$ is the displacement of the mass and $F(t)$ is an external forcing function. This equation forms the foundation for analyzing both the transient and steady-state behavior of the system.

Natural Frequency and Damping Ratio

Two key parameters fully characterize the dynamic response of a second-order system:

- The natural frequency:

$$\omega_n = \sqrt{\frac{k}{m}} \quad (7)$$

- The damping ratio:

$$\zeta = \frac{c}{2\sqrt{km}} \quad (8)$$

The natural frequency represents the rate at which the system would oscillate in the absence of damping and external forcing, while the damping ratio describes how quickly oscillations decay.

Depending on the value of ζ , the system exhibits qualitatively different behaviors:

- $\zeta = 0$: Undamped (sustained oscillations)
- $0 < \zeta < 1$: Underdamped (oscillatory decay)
- $\zeta = 1$: Critically damped (fastest non-oscillatory return)
- $\zeta > 1$: Overdamped (slow, non-oscillatory response)

In this experiment, the system is expected to be underdamped, resulting in oscillatory motion that gradually decays over time.

Transient Response

When the system is displaced from equilibrium and released, its motion is governed by the homogeneous solution to the equation of motion. For an underdamped system, the displacement takes the form:

$$y(t) = Ae^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (9)$$

where:

- $\omega_d = \omega_n \sqrt{1 - \zeta^2}$
- A and ϕ are determined by initial conditions.

This expression shows that the oscillation amplitude decays exponentially over time due to damping. In the experiment, this transient behavior can be observed when the mass is released and allowed to oscillate freely, creating a step response. By identifying two consecutive peaks in the step response (*figure 4*) to compute $\omega_d = 2\pi * (\frac{1}{t_2 - t_1})$, the damping ratio can be experimentally determined.

Forced (Steady-State) Response and Frequency Response

When the system is subjected to a harmonic forcing function of the form:

$$F(t) = F_0 \sin(\omega t + \phi) \quad (10)$$

the response consists of two components: a transient part (which decays over time) and a steady-state part that persists. At sufficiently long times, only the steady-state response remains:

$$y(t) = A_0 \sin(\omega_d t + \phi) \quad (11)$$

The amplitude A_0 depends on the forcing frequency and is given by:

$$A_0 = \frac{\frac{F_0}{k}}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}} \quad \text{Where } r = \frac{\omega}{\omega_n}. \quad (12)$$

This relationship defines the frequency response of the system. As the forcing frequency approaches the natural frequency ($r \approx 1$), the amplitude reaches a maximum, a phenomenon known as resonance. The sharpness and height of this resonance peak

depend strongly on the damping ratio: lower damping results in a higher and narrower peak.

Relevance to the Experiment

In this experiment, the ultrasonic sensor provides a direct measurement of displacement $y(t)$, enabling both transient and steady-state analyses. Transient response measurements allow determination of damping ratio and natural frequency through time-domain methods such as peak decay analysis. Steady-state response measurements across a range of input frequencies enable construction of a frequency response curve and identification of the resonance frequency. By comparing the experimentally measured response to the theoretical predictions of the second-order model, deviations can be identified. These discrepancies may arise from non-idealities such as nonlinear damping, friction in the guide mechanism, or variations in effective stiffness at larger amplitudes.

Equation Summary

Distance

$$d = \frac{c\Delta t}{2} \quad (13)$$

Where:

- c is the speed of sound
- Δt is the time between the pulse leaving and the echo returning

2nd Order Equations

Resonance Frequency

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \quad (14)$$

Where:

- ω_r is the resonant frequency
- ω_n is the natural frequency
- ζ is the damping ratio

Damped Natural Frequency

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (15)$$

Where:

- ω_d is the damped natural frequency
- ω_n is the natural frequency
- ζ is the damping ratio

2nd Order Output - Transient Response

$$y(t) = y_\infty + (y_0 - y_\infty)e^{(-\zeta\omega_n t) \left[\left(\frac{\zeta}{\sqrt{1-\zeta^2}} \right) \sin(\omega_d t) + \cos(\omega_d t) \right]} \quad (16)$$

Where:

- y_∞ is the steady state value
- y_0 is the initial value
- ω_d is the damped natural frequency
- ω_n is the natural frequency
- ζ is the damping ratio
- t is the time from the change in input

Envelope Function

$$y_{envelope}(t) = y_\infty + (y_0 - y_\infty)e^{(-\zeta\omega_n t)} \quad (17)$$

Where:

- y_∞ is the steady state value
- y_0 is the initial value
- ω_n is the natural frequency
- ζ is the damping ratio
- t is the time from the change in input

2nd Order Output - Steady State Sinusoidal Response

$$y(t) = KAM(\omega)\sin(\omega t + \phi(\omega) + \phi_0) + KA_0 + O \quad (18)$$

Where:

- K is the sensitivity
- A is the input amplitude
- $M(\omega)$ is the value of the magnitude ratio for a given frequency
- ω is the frequency
- $\phi(\omega)$ is the phase shift for a given frequency
- ϕ_0 is the initial phase shift
- A_0 is the output amplitude
- O is the output offset

Magnitude Ratio

$$M(\omega) = \frac{1}{\sqrt{(1 - (\frac{\omega}{\omega_n})^2)^2 + (\frac{2\zeta\omega}{\omega_n})^2}} \quad (19)$$

Where:

- ω is the frequency
- ω_n is the natural frequency
- ζ is the damping ratio

Phase Lag

$$\phi(\omega) = -\arctan\left(\frac{2\zeta\frac{\omega}{\omega_n}}{1 - \frac{\omega^2}{\omega_n^2}}\right) \quad (20)$$

Where:

- ω is the frequency
- ω_n is the natural frequency
- ζ is the damping ratio

Methods and Materials

Part 1: Hardware Setup and Power Supply Configuration

Before making any electrical connections, ensure that the DC power supply is turned off. Set the output voltage to +15 V by adjusting the voltage dial and confirming the value on the display before connecting any wires. Once verified, connect the +25 V terminal of the sensor's power cord to the positive (+) terminal of the power supply, and connect the ground/common (COM) lead of the sensor to the COM terminal of the supply. Carefully double-check polarity before proceeding.

After making the connections, inspect all wiring for loose contacts, damaged insulation, or potential short circuits. Ensure that all cables are routed safely away from any moving components of the oscillator apparatus. Once everything is confirmed, power on the DC supply and verify that the output is +15 V, using a multimeter if available. Do not proceed until the correct voltage is confirmed.

Part 2: Oscilloscope Connections

Begin by establishing a common ground reference across the system. Connect the ground lead of the echo signal to the COM terminal of the power supply, and ensure the ultrasonic sensor's ground is tied to this same node. The echo output should then be connected to this shared ground (see Figure 1 for a ground node diagram).

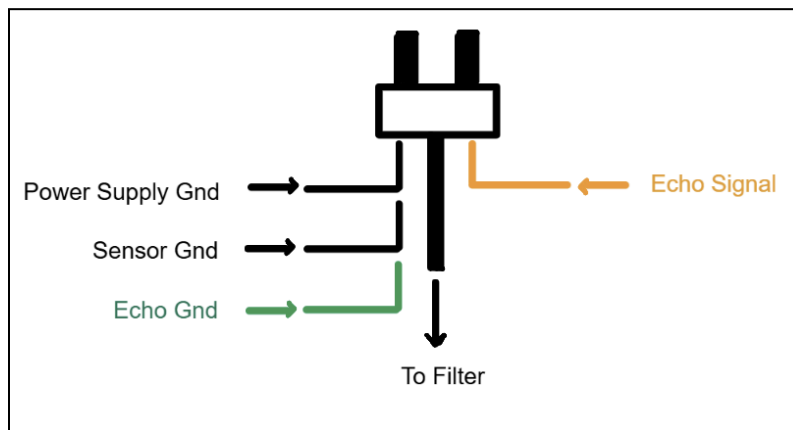


Figure 1. Echo and Ground Connections

To obtain a clean pulse and echo signal, route the sensor output through the filter. Connect the Echo/Ground output signal wire to Channel 1 input of the Butterworth Bessel dual-channel filter. Using a BNC cable, connect Channel 2 input to Channel 1 output, and then connect Channel 2 output to Channel 1 on the oscilloscope. Since the ultrasonic sensor operates around 200 kHz, set Channel 1 (high-pass filter) to a cutoff

frequency of 190 kHz and Channel 2 (low-pass filter) to 210 kHz. This configuration creates a bandpass filter that reduces noise while preserving the pulse and echo signals.

Finally, verify that the signal and ground connections are correct. Reversed connections may result in an inverted or missing signal on the oscilloscope.

Part 3: Oscilloscope Parameter Configuration

Power on the oscilloscope and configure the display settings. Set the time base to 250 μs , which provides a suitable window to capture both the transmitted pulse and its corresponding echo. Adjust the vertical scale on Channel 1 to 1.0 V to clearly visualize the signal without clipping; adjust as needed depending on signal amplitude.

Before proceeding, perform a quick spot check to ensure these settings align with the sensor's operating specifications, particularly the expected voltage range, frequency range, and pulse width. Configure the trigger settings as follows: use an edge trigger with Channel 1 as the source, select a rising edge slope, and set the mode to "Normal" so the display updates only when a valid signal is detected. Use AC coupling to remove any DC offset from the trigger signal.

Adjust the trigger level to a value just above the noise floor but below the expected peak of the echo signal. This ensures consistent triggering on the actual signal rather than background noise. Once configured, confirm that a stable, repeating waveform is visible. You should observe the initial transmitted pulse followed by a delayed echo corresponding to the target distance (see Figure 2).

Part 4: Verification and Pre-Measurement Checks

With the oscillator at rest, measure the distance between the sensor and the reflective plate. Using the oscilloscope cursors, verify that the time delay between the transmitted pulse and the echo corresponds to the expected round-trip travel time of sound (per Equation 2). When the plate is moved manually, the echo signal should shift accordingly on the display.

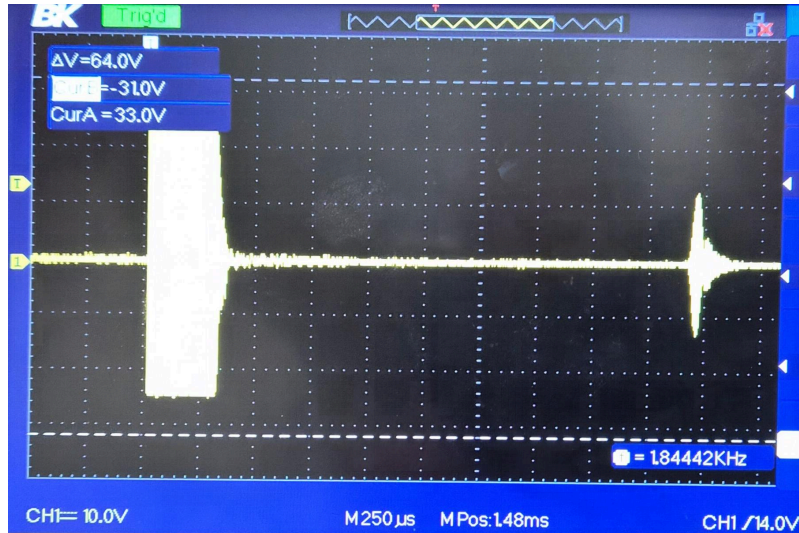


Figure 2. Pulse and Echo Visualized On Oscilloscope

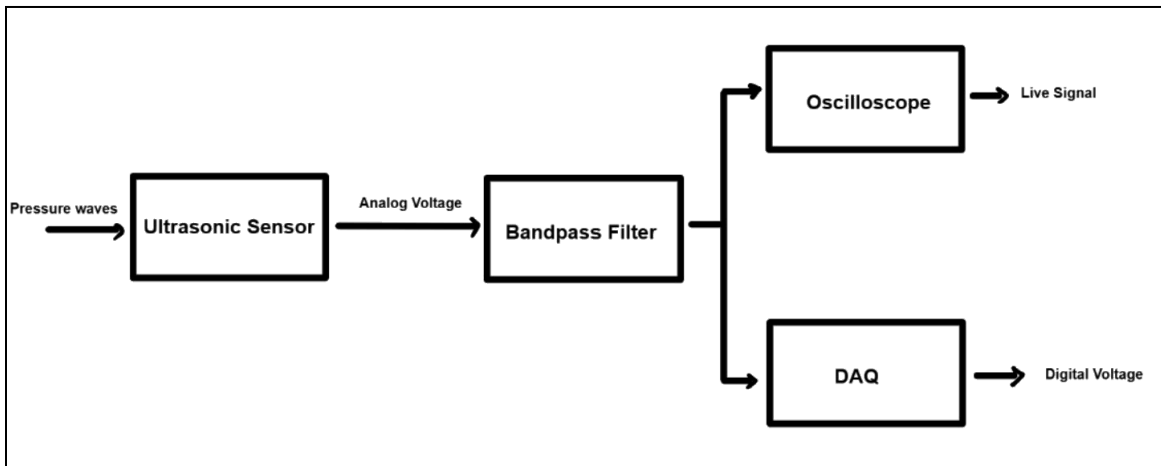


Figure 3. Experiment Block Diagram

Part 5: Data Acquisition

To begin data acquisition, connect Channel 1 of the oscilloscope to Channel 1 of the DAQ board using a T-splitter so the signal is shared between both devices. Open the provided MATLAB script (NIDAQ_template_1Channel.m) and run it to observe the initial output. The script will require modification to fully analyze the data, as the goal is to convert the time delay between the transmitted pulse and echo into a distance measurement. When the oscillator is operating, the resulting displacement signal should appear sinusoidal.

Next, determine the spring constant using static deflection without removing the spring from the shaft. Record the mass of the top collar as labeled (in grams), and do not remove it, as doing so may disrupt internal components and alter damping behavior. Measure the

natural frequency of the upper mass using two different methods, record both values, and compare them. Repeat this process to determine the damping ratio using two independent methods, again noting any discrepancies.

Confirm that the ultrasonic sensor is producing stable and consistent distance readings, and note any DC offset present in the signal for later analysis. Set the motor to a specific frequency and determine the corresponding driving frequency in Hz, recording both the knob setting and frequency. Measure the total linear displacement of the lower mass (peak-to-peak input amplitude), as this will be used in frequency response analysis.

With the system running, measure the displacement of the upper mass as a function of time, recording both the peak amplitude and oscillation frequency. Determine the minimum sampling rate required to avoid aliasing, ensuring it satisfies the Nyquist criterion.

Finally, measure the frequency response of the system by repeating these measurements across a range of driving frequencies. Collect data for both increasing and decreasing frequencies, using at least 25 data points to generate a well-resolved response curve. Ensure the range spans below and above the resonance frequency. At each step, allow the system to reach steady state before recording data, as transient behavior will distort the results.

Equipment List

Equipment	Manufacturer	Model Number	Resolution	Voltage/current	Error Parameters
Ultrasonic Sensor	AirMar	AT-200	1mm	Powered at +15V DC	n/a
DAQ Board	National Instruments	BNC 2090	16-bit	$\pm 10V$ input, 250 kS/s max	n/a
DC Power Supply	Hewlett Packard	E3631A	10mV / 10mA	0-25V / 0-1A	n/a
Oscilloscope	BK Precision	2190E	8-bit vertical	$\pm 5V$	n/a
Filter	Krohn-Hite	3940	n/a	$\pm 15V$ supply	n/a
Driving Motor	Bison	011-190-0010	n/a	90V DC, 0.62A max	n/a

Spring	n/a	n/a	n/a	n/a	n/a
Mass/Plate	n/a	n/a	1mm	n/a	n/a
Ruler	Faber-Castell	17 70 91	1mm	n/a	$\pm 0.5\text{mm}$

Table 1. Equipment List for all instruments used

Analysis & Uncertainty Analysis

Spring Constant Calculation:

The spring constant, k , is a measure of a spring's stiffness and describes the relationship between the force applied to a spring and the resulting deformation. It is a fundamental parameter of any spring-mass system and is required to theoretically predict our system's dynamic behavior.

The spring constant is derived from Hooke's Law, which states that the force exerted by a spring is linearly proportional to its displacement from its natural (unloaded) length:

$$F = k \times \delta \quad \text{OR} \quad k = F / \delta$$

Where: F is the applied force in Newtons, k is the spring constant in N/m , and δ is the displacement of the spring from its equilibrium position in meters. In this experiment, the spring constant was determined using the static deflection method. A series of known masses ranging from 200g to 1500g were placed on the upper collar of the oscillator, and the resulting deflection of the spring was measured at each load. The applied force at each loading step was calculated from the weight of the added mass: $F = m \times g$. The deflection δ was determined by measuring the spring height at each load and subtracting it from the baseline spring height measured with only the collar and plate and no added mass:

$\delta = H_0 - H_1$ where H_0 is the equilibrium spring height with no added mass (28.9 cm) and H_1 is the spring height under each respective load. All height measurements were taken in centimeters and converted to meters for consistency with SI units. A spring constant value was calculated for each loading step using $k = F/\delta$, and the final spring constant was taken as the average across all loading steps. It is worth noting that the spring exhibited a progressive increase in k with increasing load, particularly at higher masses of 1000g and above. This behavior is characteristic of a weak nonlinear spring, where the linear assumption of Hooke's Law begins to break down at larger deformations. This is consistent with the nature of the experiment as described in the project manual, which notes that all real mechanical systems contain some inherent nonlinearity. Despite this, the static deflection method provides a reasonable first-order estimate of k across the loading range, and the average value accounts for both the linear and weakly nonlinear behavior observed across all measurements.

Full Height (cm)	Offset Height Above Spring (cm)	Offset Height (cm)	Total Offset Height (cm)	Equilibrium Spring Height (cm)	Equilibrium Spring Height (m)
50.3	4.4	17	21.4	28.9	0.289

Table 2. System Baseline Measurements for K-value calculations

Mass (g)	Mass (kg)	Force (N)	Full Height (cm)	Spring Height (cm)	Spring Height (m)	δ (m)	k (N/m)
200	0.2	1.962	45.5	24.1	0.241	0.048	40.88
300	0.3	2.943	44.5	23.1	0.231	0.058	50.74
400	0.4	3.924	42.9	21.5	0.215	0.074	53.03
500	0.5	4.905	42.4	21.0	0.210	0.079	62.09
1000	1.0	9.810	35.2	13.8	0.138	0.151	64.97
1500	1.5	14.715	31.0	9.6	0.096	0.193	76.24

Table 3. Static Deflection Data and Spring Constant Calculation ($k = F/\delta$)

The mean spring constant was found to be $k = 57.99 \text{ N/m}$, which was subsequently used to calculate the theoretical natural frequency of the system.

Spring Constant Uncertainty

The spring constant contains both systematic uncertainty and random uncertainty. The random uncertainty can be calculated by applying a 95% confidence t-distribution. Where $N = 6$ and the t value is equal to 2.447 acquired from the t-distribution table.

$$u_{P,k} = \pm t_{95\%} \cdot \frac{s_x}{\sqrt{N}} \text{ where } s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$u_{P,k} = 2.447 * \frac{12.4 \frac{N}{m}}{\sqrt{6}} = \pm 12.39 \frac{N}{m}$$

The random uncertainty of our spring constant measurement will be far greater than our systematic uncertainty. Given that the uncertainty is already very large, we can take the random uncertainty to be our total uncertainty. So the uncertainty in our spring constant will be $\pm 12.39 \frac{N}{m}$.

Natural Frequency Calculations

Method 1:

The natural frequency of the upper mass was first determined theoretically using the measured system parameters. The natural frequency of a spring-mass system is defined as:

$$\omega_n = \sqrt{\frac{k}{m}}$$

Where k is the spring constant in N/m and m is the total oscillating mass in kg. The total mass of the system was determined by summing the collar mass and the plate mass, both of which were measured directly. The collar mass was read from the value inscribed on the collar itself, and the combined mass of the collar and plate was found to be 114.6 g, or 0.1146 kg. Using the experimentally determined average spring constant of $k = 57.99$ N/m, the theoretical natural frequency was calculated as:

$$\omega_n = \sqrt{\frac{57.99}{0.1146}}$$

$$\omega_n = 22.495 \frac{rad}{s}$$

This was then converted to Hz using:

$$f_n = \frac{\omega_n}{2\pi}$$

$$f_n = \frac{22.496}{2\pi}$$

$$f_n = 3.582 \text{ Hz}$$

This theoretical value represents the frequency at which the upper mass is expected to naturally oscillate when displaced from its equilibrium position and released, based purely on the physical properties of the system.

ω_n [Hz]	22.495
f_n [Hz]	3.582

Table 4. Natural frequency from k (Method 1)

Uncertainty in Natural Frequency: Method 1

To find the uncertainty of the natural frequency using Method 1, the uncertainty in our spring constant needs to be propagated through the natural frequency equation. The mass will not affect our uncertainty as this is a given constant written on the collar.

$$\omega_n = \sqrt{\frac{k}{m}}$$

To propagate the uncertainty we need take the partial derivative of the natural frequency with respect to the spring constant

$$U_{\omega_n} = \frac{\partial \omega_n}{\partial k} (u_k)$$

$$\frac{\partial \omega_n}{\partial k} = \frac{1}{2\sqrt{km}}$$

The uncertainty of our spring constant was calculated before and found to be $u_k = 12.39 \frac{N}{m}$.

$$U_{\omega_n} = \frac{1}{2\sqrt{km}} (u_k) = \frac{1}{2\sqrt{57.99(\frac{N}{m}) * 0.1146kg}} (12.39 \frac{N}{m})$$

$$U_{\omega_n,1} = \pm 2.403 \frac{rad}{s}$$

Method 2

To determine the natural frequency using a second method, the resonance frequency ω_r and damping ratio ζ is used in the equation

$$\omega_n = \frac{\omega_r}{\sqrt{1-2\zeta^2}}$$

The resonance frequency ω_r was identified experimentally by examining the amplitude of the top mass oscillations across all knob settings in Table 2. The knob setting that produced the largest peak-to-peak displacement in the Distance vs Time plots in Figure 6 was knob_44, indicating

that the system was operating at or near resonance at that setting. The driving frequency at knob_44 was recorded for increasing frequency sweep to be $\omega_r = 15.733 \frac{rad}{s}$. The damping ratio used here is $\zeta = 0.107$, as determined by the peak magnitude method detailed in Damping Ratio Method 2. Using the experimentally determined resonance frequency ω_r and damping ratio ζ , the natural frequency was calculated using the relationship between resonance frequency and natural frequency for a damped second order system:

$$\omega_n = \frac{\omega_r}{\sqrt{1-2\zeta^2}} = \frac{15.733 \frac{rad}{s}}{\sqrt{1-2(0.107)^2}} = 15.916 \frac{rad}{s}$$

To convert to Hz simply use the following equation.

$$f_n = \frac{\omega_n}{2\pi}$$

Uncertainty in Natural Frequency: Method 2

The second method used to calculate the natural frequency utilized the values of the resonance frequency and damping ratio. These values have uncertainties that we have previously solved for. The uncertainty of the resonance frequency is found in *table 2* to be $U_{\omega_r} = 0.93 \frac{rad}{s}$ when converting to rad/s, and the damping ratio from Method 2 has an uncertainty of $U_{\xi,2} = \pm 0.003$. These uncertainties propagate through the natural frequency equation used in Method 2.

$$\omega_n = \frac{\omega_r}{\sqrt{1-2\zeta^2}}$$

To propagate the uncertainty of the resonance frequency and the damping ratio, the following equation must be used.

$$U_{\omega_n,2} = \sqrt{\left(\frac{\partial \omega_n}{\partial \omega_r} U_{\omega_r}\right)^2 + \left(\frac{\partial \omega_n}{\partial \xi} U_{\xi,2}\right)^2}$$

The partial derivatives of the natural frequency with respect to the resonance frequency and damping ratios are as follows:

$$\frac{\partial \omega_n}{\partial \omega_r} = \frac{1}{\sqrt{1-2\xi^2}}, \quad \frac{\partial \omega_n}{\partial \xi} = \frac{2\xi \omega_r}{(1-2\xi^2)^{\frac{3}{2}}}$$

Plugging in for the corresponding damping frequency found using method 2, $\xi = 0.107$ and natural frequency of Method 2, $\omega_n = 15.916$ rad/s, the uncertainty of the natural frequency using Method 2 is found to be

$$U_{\omega_n, 2} = 0.941 \frac{\text{rad}}{\text{s}}$$

	METHOD 1	METHOD 2
	$\omega_n = \sqrt{\frac{k}{m}}$	$\omega_n = \frac{\omega_r}{\sqrt{1-2\xi^2}}$
ω_n (rad/s)	22.495	15.916
f_n (Hz)	3.582	2.533

Table 5. Natural Frequency Methods 1 and 2

Damping Ratio Calculations

Method 1

The first method for determining the damping ratio used the logarithmic decrement method applied to the free oscillation response of the system. The top mass was manually displaced and released with the motor off, and the resulting decaying oscillation was recorded. Two amplitude peaks and the steady state offset were identified from the signal in *Figure 4* and used in the following equations:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

$$\delta = \ln\left(\frac{y_2 - y_\infty}{y_1 - y_\infty}\right)$$

Parameter	Value
y_2 (First Peak) (m)	0.478828
y_1 (Second Peak) (m)	0.43561

y_{∞} (Steady State) (m)	0.409582
δ (Logarithmic Decrement)	0.9785
ζ (Damping ratio)	0.154

Table 6. Damping ratio via Logarithmic Decrement

This method relies on two peaks to characterize the decay rate. The damping ratio obtained from this method is $\zeta = 0.154$.

Uncertainty in Damping Ratio: Method 1

Finding the uncertainty in the damping ratio calculated via method 1 requires the equations listed below.

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\zeta = \frac{c}{c_c}$$

$$c_c = 2\sqrt{km}$$

We already know that the sampling rate is

$$F_s = 50000 \frac{\text{samples}}{\text{sec}}$$

So the discretization error can be found as follows

$$F_{s, \text{uncertainty}} = \pm \frac{\Delta t}{2} = \frac{2 \cdot 10^{-5} \text{ s}}{2} = 1 \cdot 10^{-5} \text{ s}$$

Next we use the sample uncertainty to find the total distance uncertainty

$$u_{\text{sample}} = 1 \cdot 10^{-5} \text{ s} = \pm 10 \mu\text{s}$$

$$u_{total,d} = \sqrt{\left(\frac{\partial d}{\partial t} * u_{sample}\right)^2}$$

Where

$$\frac{\partial d}{\partial t} * u_{sample} = \frac{10\mu s(343\frac{m}{s})}{2}$$

Therefore

$$u_{total,distance} = \pm 1.715 * 10^{-3} m$$

Now the uncertainty for the log decrement can be properly calculated.

$$\ln\left(\frac{y_2 - y_\infty}{y_1 - y_\infty}\right) = \ln(y_2 - y_\infty) - \ln(y_1 - y_\infty)$$

$$u_{total,\delta} = u_{total,distance} \sqrt{\left(\frac{\partial \delta}{\partial y_1} u_{total,d}\right)^2 + \left(\frac{\partial \delta}{\partial 2} u_{total,d}\right)^2 + \left(\frac{\partial \delta}{\partial y_1} u_{total,d}\right)^2}$$

And

$$\frac{\partial \delta}{\partial y_2} = \frac{1}{y_2 - y_\infty} \quad \frac{\partial \delta}{\partial y_1} = \frac{1}{y_1 - y_\infty} \quad \frac{\partial \delta}{\partial y_\infty} = \frac{1}{y_1 - y_\infty} - \frac{1}{y_2 - y_\infty}$$

Putting these components of the uncertainty together using the values from Table 7 yield

$$u_{total,\delta} = \pm 0.081523m$$

Now it's possible to find the uncertainty in ζ where

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

Propagating everything to the correct units, in this case unitless, requires only a multiplication from here. Using the partial derivative of ζ with respect to delta

$$\frac{\partial \zeta}{\partial \delta} = \frac{4\pi^2}{(4\pi^2 + \delta^2)^{\frac{3}{2}}}$$

Allows us to plug in our previously calculated uncertainty in distance

$$u_{\zeta, method 1} = \frac{4\pi^2}{(4\pi^2 + \delta^2)^{\frac{3}{2}}} (u_\delta)$$

$$= \frac{4\pi^2}{(4\pi^2 + (0.9785 \text{ m})^2)^{\frac{3}{2}}} (0.08125 \text{ m})$$

$$u_{\zeta, \text{method 1}} = \pm 0.0125$$

Since this method has a higher uncertainty than method 2, we did not use this damping ratio for any other calculations in this report. This higher uncertainty is likely carry-over from the fact that our k value was non-linear over the ranges tested.

Method 2

The damping ratio was determined a second time using the peak magnitude method. For a second-order underdamped system, the peak magnitude ratio (M_{peak}) at resonance is related to the damping ratio by:

$$M_{peak} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

Rearranging to solve for ζ :

$$\zeta = \sqrt{\frac{1 - \sqrt{1 - \frac{1}{M_{peak}^2}}}{2}}$$

(M_{peak}) was identified from the experimental frequency response data in Figure 9. The maximum amplitude ratio recorded was $M_{peak} = 4.707843$. Substituting into Equation 25:

$$\zeta = \sqrt{\frac{1 - \sqrt{1 - \frac{1}{4.707843^2}}}{2}}$$

$$\zeta = 0.107$$

The damping ratio obtained from the peak magnitude method is $\zeta = 0.107$.

Uncertainty in Damping Ratio: Method 2

Finding the uncertainty in the damping ratio calculated via method 2 is much simpler than method 1. Since the damping ratio was calculated using

$$\zeta = \sqrt{\frac{1 - \sqrt{1 - \frac{1}{M_{peak}^2}}}{2}}$$

All that's needed is to take the partial derivative of with respect to M

$$\frac{\partial \zeta}{\partial M} = \frac{1}{(2\sqrt{2}(M_{peak})^{3/2} \sqrt{M_{peak} - \sqrt{M_{peak}^2 - 1}} \cdot \sqrt{M_{peak}^2 - 1}}$$

Plugging in the ascending value for $M_{peak} = 4.708$ (knob 44) and multiplying by the uncertainty in the magnitude ratio gives

$$\begin{aligned} u_{\zeta, method 2} &= u_m \cdot \frac{\partial \zeta}{\partial M} \\ &= (0.077) \left(\frac{1}{(2\sqrt{2}(4.708)^{3/2} \sqrt{4.708 - \sqrt{4.708^2 - 1}} \cdot \sqrt{4.708^2 - 1}} \right) \\ u_{\zeta, method 2} &= \pm 0.003 \end{aligned}$$

Since this method resulted in a lower uncertainty value, this is the damping ratio used throughout the rest of the report.

Driving Frequency Calculations

The driving frequencies for each knob setting were found using MatLab. The code looked at the distance vs time data for a given setting, identified the peaks and then found the time difference between them. Once every peak was accounted for, the code took an average of the frequency for that setting that can be seen in the tables below.

Knob Setting	Driving Frequency (Hz)	Uncertainty ($\pm$ Hz)
36	1.9552	0.095
38	2.1200	0.123
40	2.2678	0.127
42	2.3710	0.112
44	2.5040	0.148
46	2.6304	0.147
48	2.8093	0.191
50	2.9454	0.187
52	3.0963	0.217

54	3.2343	0.217
56	3.3666	0.249
58	3.5486	0.280
60	3.6694	0.282

Table 7. Driving Frequency for Given Knob Setting(Ascending)

Knob Setting	Driving Frequency (Hz)	Uncertainty ($\pm$ Hz)
58	3.5212	0.277
56	3.4394	0.264
54	3.2355	0.217
52	3.132	0.221
50	2.9469	0.187
48	2.7813	0.179
46	2.6304	0.145
44	2.526	0.149
42	2.4281	0.171
40	2.2788	0.125
38	2.1223	0.108
36	1.9743	0.093

Table 8. Driving Frequency for Given Knob Setting(Descending)

Driving Frequency Uncertainty

Each frequency has its own degree of uncertainty. Higher knob settings resulted in faster oscillations and more observable periods within the fixed sampling time. All uncertainties were calculated as follows, only the sample calculation for the 36 knob setting is shown.

The systematic uncertainty for the frequency was found above to be ± 20 ms. The random uncertainty can be found via

$$u_{P,x} = \pm t_{95\%} \cdot \frac{s_x}{\sqrt{N}} \text{ where } s_x = \sqrt{\frac{\sum (X_i - \widehat{X})^2}{n-1}}$$

N varies based on the driving frequency, this means the degrees of freedom for the Student's T table also changes. For the knob setting corresponding to 36 on the dial; the standard deviation was 0.0192 sec, the number of cycles was 9 meaning the degrees of freedom is 8, giving a student's t table value of 2.306.

Thus a sample calculation for this settings frequency uncertainty can be calculated as:

$$u_{P,x} = (2.306) \frac{0.0192 \text{ sec}}{\sqrt{9}} = \pm 0.0147 \text{ sec}$$

Combining these uncertainties is possible if they are propagated into the correct units via

$$u_{tot, freq} = \sqrt{\left(\frac{\partial f}{\partial x} U_x\right)^2 + \left(\frac{\partial f}{\partial y} U_y\right)^2}$$

Here we chose to convert everything into Hz. So the function the partial derivative is being applied to is

$$f = \frac{1}{T} \rightarrow \frac{\partial f}{\partial x} = \frac{1}{T^2}$$

Here you can see that the period of oscillation impacts the systematic uncertainty as well as the random uncertainty indirectly. Plugging in the values from the sample calculation

$$u_{tot, knob 36} = \sqrt{\left(\frac{1}{(0.5114 \text{ sec})^2} \cdot 20 \text{ ms}\right)^2 + \left(\frac{1}{(0.5114 \text{ sec})^2} \cdot 0.0147 \text{ sec}\right)^2} = 0.095 \text{ Hz}$$

The uncertainties for tables 2 and 3 of this report were calculated by applying this same method to each individual frequency.

Damping Coefficient Calculation

The damping coefficient c appears in the equation for the damping ratio

$$\zeta = \frac{c}{2\sqrt{km}}$$

Since the other values are known: $\zeta = 0.107$, $k = 57.99 \text{ N/m}$, $m = 0.1146 \text{ kg}$ it's a simple calculation to solve this equation for c

$$\begin{aligned}
c &= 2\zeta\sqrt{km} \\
&= 2(0.107)\sqrt{(57.99 \text{ N/m})(0.1146 \text{ kg})} \\
&= 0.5516 \frac{\text{N}\cdot\text{m}}{\text{s}}
\end{aligned}$$

Damping Coefficient Uncertainty

Since m was given on the collar, the associated uncertainty can be found by

$$u_c = \sqrt{\left(\frac{\partial c}{\partial \zeta} \cdot u_\zeta\right)^2 + \left(\frac{\partial c}{\partial k} \cdot u_k\right)^2}$$

The uncertainties for ζ and k were found above so all that's needed is to plug in their values

$$\begin{aligned}
u_c &= \sqrt{(2\sqrt{km} \cdot u_\zeta)^2 + (\zeta\sqrt{\frac{m}{k}} \cdot u_k)^2} \\
&= \sqrt{(2\sqrt{(57.99 \text{ N/m})(0.1146 \text{ kg})} \cdot 0.003)^2 + ((0.107)\sqrt{\frac{0.1146 \text{ kg}}{57.99 \text{ N/m}}} \cdot 12.31 \text{ N/m})^2} \\
u_c &= \pm 0.0606 \frac{\text{N}\cdot\text{m}}{\text{s}}
\end{aligned}$$

Other Uncertainty Analysis

The uncertainty of our measurements, primarily the frequency of oscillation and maximum displacement are governed by the following uncertainty .

DAQ Sampling Rate

The DAQ sampling rate is the factor that governs our uncertainty in our distance measurements as this is the resolution that accurately determines our distance calculation. To find its contribution, we must first calculate the time elapsed between each sample:

$$\Delta t = \frac{1}{f_s} = \frac{1}{50,000} = 20 \mu\text{s}$$

Thus

$$u_{t, DAQ} = \pm \frac{\Delta t}{2} = \pm 10 \mu\text{s}$$

This uncertainty is propagated through our distance calculation which is dependent on the following equation

$$d = \frac{c\Delta t}{2}$$

In order to propagate our error to our displacement measurements we need to find the partial derivative with respect to the time difference.

$$\frac{\delta d}{\delta t} = \frac{c}{2}$$

The propagation is then calculated by

$$U_d = \frac{\partial d}{\partial t} * u_{t,DAQ} = \frac{343(\frac{m}{s})(10\mu s)}{2} = \pm 1.715 * 10^{-3} m$$

When calculating the maximum displacement of our system we found that the uncertainty of the measurement was dependent on the random and systematic uncertainty. The systematic uncertainty has already been calculated, however the random uncertainty still needs to be found. To find the maximum displacement we found the difference between one peak and one trough in our distance vs. time graph.

$$\Delta x_{max} = x_{peak} - x_{trough}$$

$$\Delta x_{max} = 0.493m - 0.275m = 0.218m$$

This was found for multiple cycles in our collected data. Each knob setting had a different frequency, and as a result, each setting had a different number of cycles resulting in varied random uncertainty. Therefore, each maximum displacement has different uncertainty. The random uncertainty for ascending knob_setting 60 was found. The random uncertainty was for 95% confidence T distribution using the FigBeasley Table. The number of cycles in this measurement was N=18 and the corresponding T value was 2.110. To find the random uncertainty the following equation was used

$$u_{P,d} = \pm t_{95\%} \cdot \frac{s_x}{\sqrt{N}} \text{ where } s_x = \sqrt{\frac{\sum (X_i - \hat{X}_i)^2}{n-1}}$$

The mean maximum displacement was calculated to be 0.0575m and the corresponding standard deviation was 0.001967m .

$$u_{P,d} = (2.110) \frac{0.001967m}{\sqrt{18}} = 9.782 * 10^{-4} m$$

To calculate the total uncertainty the systematic and random uncertainty must be combined with the following equation:

$$U_{total,d} = \sqrt{u_{P,d}^2 + U_d^2} = \sqrt{0.000978m^2 + .001715m^2} = 0.001974 m$$

The mean maximum displacement for ascending or descending knob values and their corresponding uncertainties are found in tables 9 and 10.

Knob Setting	Maximum Mean Displacement (m)	Uncertainty($\pm$ m)
36	0.1335	0.00216
38	0.1608	0.00188
40	0.21609	0.00234
42	0.24519	0.00256
44	0.27451	0.00249
46	0.24474	0.00189
48	0.19683	0.00218
50	0.14553	0.00229
52	0.11735	0.00236
54	0.09453	0.00204
56	0.07966	0.00185
58	0.06618	0.00212
60	0.05755	0.00197

Table 9. Mean Maximum Displacement and Uncertainty for Corresponding Ascending Knob Setting

Knob Setting	Maximum Mean Displacement (m)	Uncertainty($\pm$ m)
58	0.06634	0.00192
56	0.0782	0.00188
54	0.09342	0.00188

52	0.11173	0.00196
50	0.14602	0.00241
48	0.20126	0.00283
46	0.24484	0.0022
44	0.27589	0.00244
42	0.24453	0.0021
40	0.2106	0.00221
38	0.1678	0.00205
36	0.1369	0.00198

Table 10. Mean Maximum Displacement and Uncertainty for Corresponding Descending Knob Setting

The bottom mass maximum displacement was calculated in the same manner as the top mass displacements. To accomplish this we set the knob to the lowest setting in order to achieve an extremely low frequency. When frequency is low, the displacement of the bottom mass is equal to the displacement of the top mass. This was done for one cycle in order to obtain the bottom mass maximum displacement which can be seen in table 11.

Bottom Mass Maximum Displacement (m)	Uncertainty ($\pm m$)
0.05831	0.001715

Table 11. Bottom Mass Maximum Displacement

Ultrasonic Time Uncertainty

The Ultrasonic Sensor sends and receives pulses at a rate of 25 Hz or 25 samples per second. This uncertainty governs the uncertainty in the frequencies we measure. The standard uncertainty for these discrete time points is

$$u_{t,ult} = \frac{\Delta t}{2} = \frac{40 \text{ ms}}{2} = \pm 20 \text{ ms}$$

This uncertainty is what dictates the uncertainty in the frequency.

Magnitude Ratio Uncertainty

The equation for M the magnitude ratio is

$$M = \frac{\text{output}}{\text{input}} = \frac{X_{\text{top mass}}}{X_{\text{bottom mass}}}$$

The systematic uncertainty is just the same as the uncertainty in distance found above, 1.715×10^{-3} m.

The random uncertainty can be found by taking the proper partial derivatives, multiplying them by the respective distance uncertainties, and adding the sum in quadrature.

$$u_m = \sqrt{\left(\frac{\partial M}{\partial x_{\text{top}}} \cdot u_{x \text{ top}}\right)^2 + \left(\frac{\partial M}{\partial x_{\text{bot}}} \cdot u_{x \text{ bot}}\right)^2}$$

Since the uncertainty in each measurement is different, a sample calculation for the uncertainty corresponding to a knob setting of 36 is shown below. $X_{\text{top}} = 0.1335$ m, $X_{\text{bot}} = 0.05831$ m, with the uncertainties being ± 0.002 m and ± 0.0017 m respectively.

Plugging in these values yields

$$M = \frac{0.1335 \text{ m}}{0.05831 \text{ m}} = 2.2894$$

Thus u_m can be found to be

$$\begin{aligned} u_m &= \sqrt{\left(\frac{1}{x_{\text{bot}}} \cdot u_{x \text{ top}}\right)^2 + \left(\frac{x_{\text{top}}}{x_{\text{bot}}^2} \cdot u_{x \text{ bot}}\right)^2} \\ &= \sqrt{\left(\frac{1}{0.0531 \text{ m}} \cdot 0.1335 \text{ m}\right)^2 + \left(\frac{0.1335 \text{ m}}{(0.0531 \text{ m})^2} \cdot 0.0017 \text{ m}\right)^2} \\ &= \pm 0.077 \end{aligned}$$

The following tables contain the values of ascending and descending magnitude ratios and their corresponding uncertainties.

Knob Setting	Magnitude Ratio	Uncertainty
36	2.2894	0.0768
38	2.7576	0.0873
40	3.7059	0.1162
42	4.2049	0.1312
44	4.7078	0.1449
46	4.1973	0.1276
48	3.3756	0.1061
50	2.4958	0.0832
52	2.0125	0.0717
54	1.6212	0.0591
56	1.3662	0.0512
58	1.1349	0.0494
60	0.9869	0.0446

Table 12. Magnitude Ratio and Uncertainty for Corresponding Ascending Knob Setting

Knob Setting	Magnitude Ratio	Uncertainty
58	1.1377	0.0469
56	1.3412	0.0509
54	1.6022	0.0571
52	1.9161	0.0656
50	2.5042	0.0845
48	3.4516	0.1125
46	4.199	0.1291
44	4.7314	0.1453

42	4.1936	0.1285
40	3.6118	0.1128
38	2.8776	0.0916
36	2.3477	0.077

Table 13. Magnitude Ratio and Uncertainty for Corresponding Descending Knob Setting

Results

Table 14 consolidates all experimentally determined system parameters and their associated uncertainties, summarizing the key dynamic characteristics of the spring-mass-damper system as obtained through the methods described in the Analysis section.

System Characteristics	System Values	Uncertainty
$k \left[\frac{N}{m} \right]$	57.99	$\pm 12.39 \frac{N}{m}$
$\omega_n \left[\frac{rad}{s} \right]$	Method 1: 22.495 Method 2: 15.916	$\pm 0.941 \frac{rad}{s}$ (Method 2)
ζ [unitless]	Method 1: 0.154 Method 2: 0.107	± 0.003 (Method 2)
$c = \zeta(2\sqrt{km}) \left[\frac{N*s}{m} \right]$	0.5516	± 0.0606
M [kg]	0.1146 kg	N/A (Given Value)

Table 14. System Characteristics and their uncertainty

Figure 4 presents the displacement of the upper mass as a function of time following a manual step displacement with the motor off. The decaying oscillatory behavior is characteristic of an underdamped second-order system, and the two labeled peaks were used to apply the logarithmic decrement method for determining the damping ratio for our method 1. The steady-state value to which the signal converges corresponds to the equilibrium position of the upper mass and was used as the reference offset.

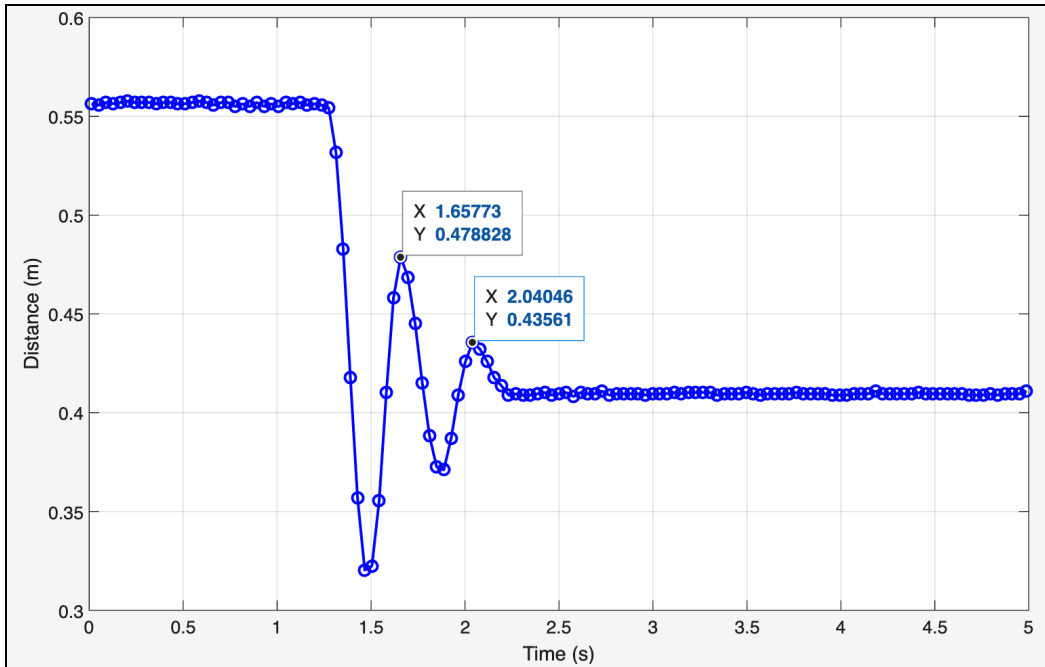


Figure 4. Damped Response To Step Input

Figure 5 displays the raw analog voltage signal acquired by the DAQ board from the ultrasonic sensor. The large periodic voltage spikes correspond to the transmitted pulse and the spike that appears shortly after is its returning echo, with the time delay between them proportional to the distance between the sensor and the reflective collar.

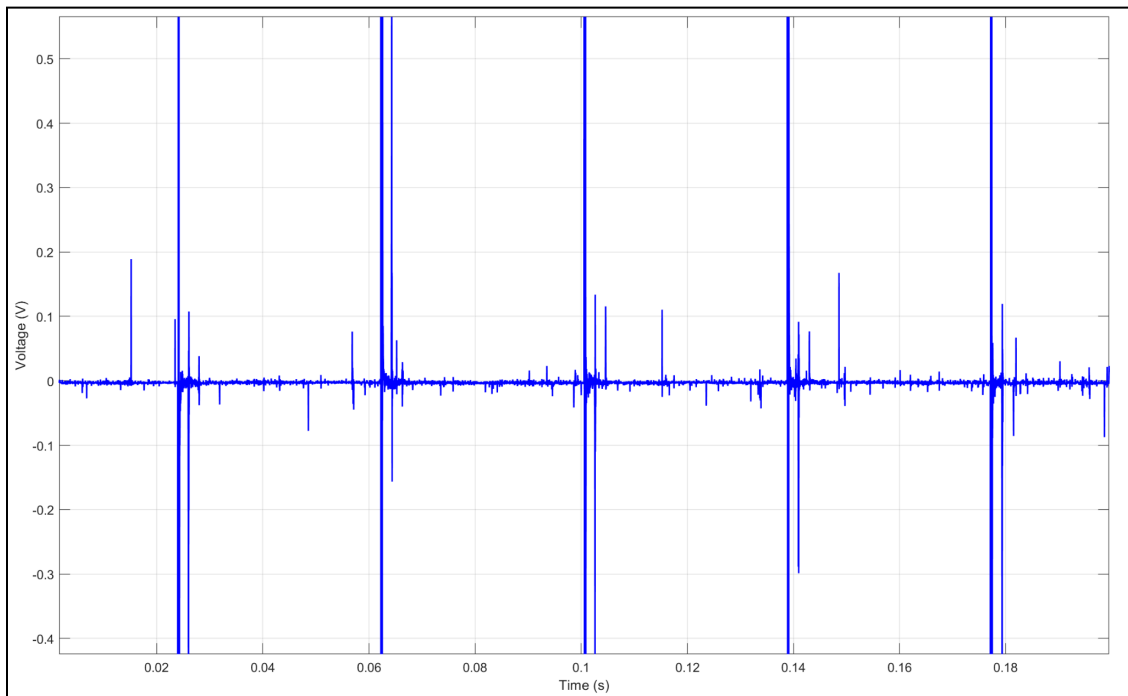


Figure 5. Raw Voltage vs Time Data

Figure 6 shows the processed displacement of the upper mass as a function of time for each knob setting in the ascending frequency sweep, arranged from the lowest to the highest driving frequency. As the knob setting increases toward the resonance condition at knob 44, the peak-to-peak amplitude of the oscillation grows noticeably before diminishing at higher settings, consistent with the expected behavior of a driven second-order system approaching and then passing through resonance.

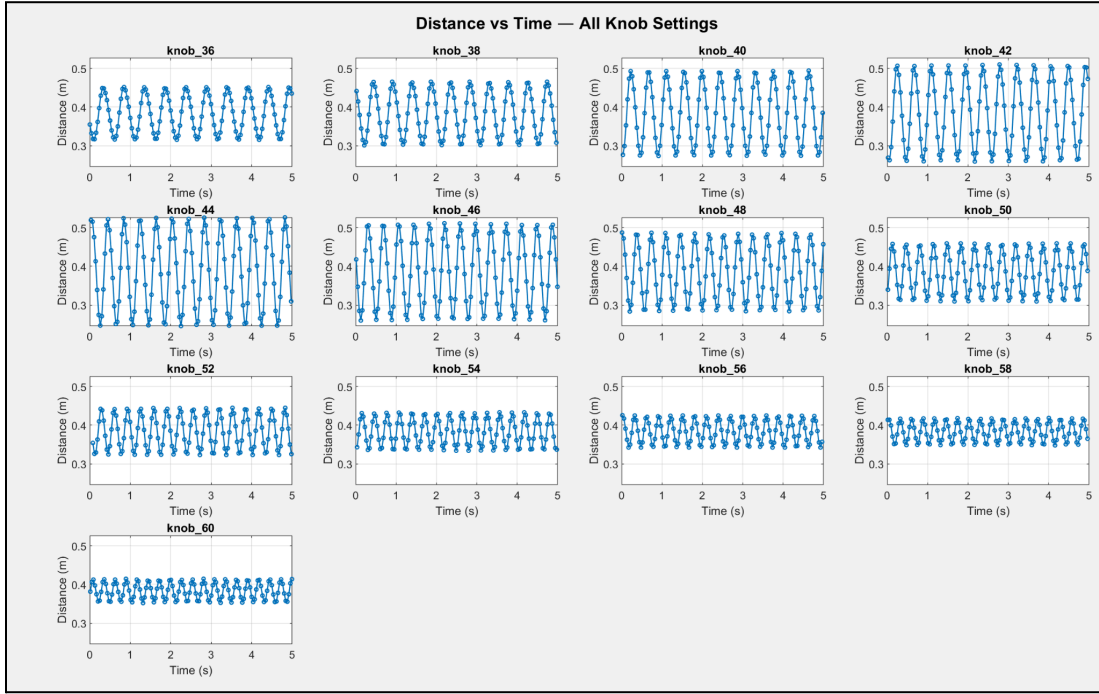


Figure 6. Distance vs Time for Different Knob Settings(Ascending)

Figure 7 presents the processed displacement of the upper mass as a function of time for each knob setting in the descending frequency sweep, in which the motor was stepped down from the highest to the lowest driving frequency. As with the ascending sweep, the peak-to-peak amplitude reaches its maximum at knob setting 44 before diminishing in both directions, confirming the consistency of the resonance identification across both sweep directions.

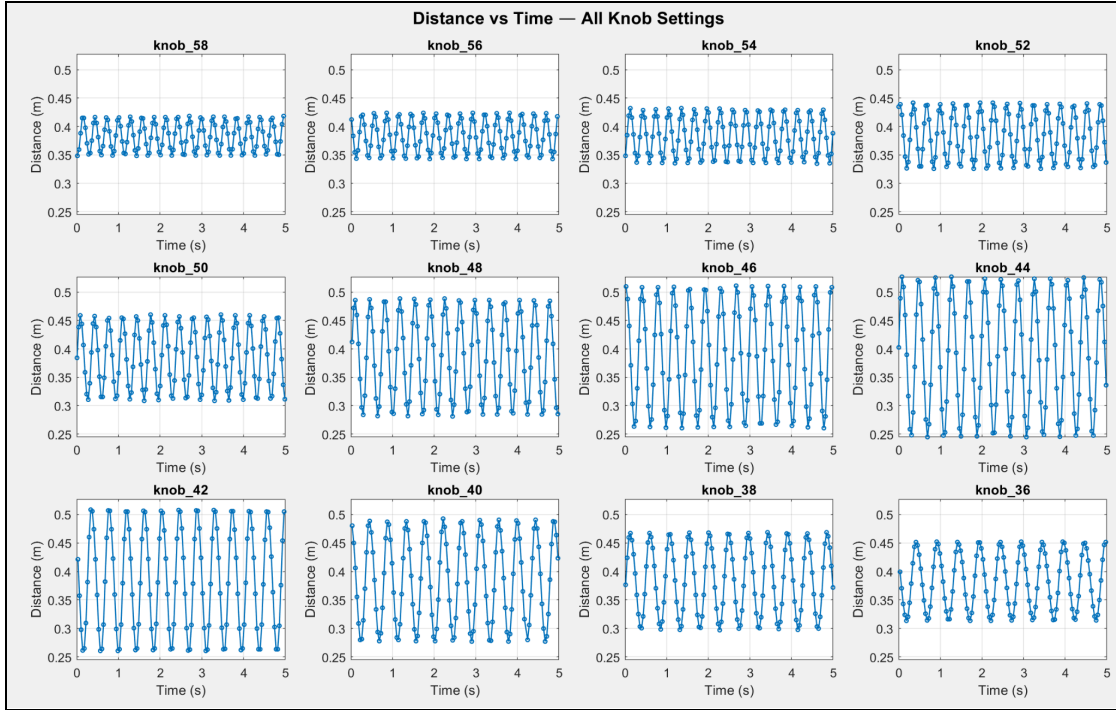


Figure 7. Distance vs Time for Different Knob Settings(Descending)

Comparing Figures 6 and 7 at equivalent knob settings reveals a high degree of agreement between the ascending and descending sweeps, with both exhibiting their maximum oscillation amplitude at knob setting 44 and displaying clean, sinusoidal waveforms throughout the frequency range. The amplitude values at resonance were 0.2745 m and 0.2759 m for the ascending and descending sweeps respectively, a difference of less than 0.6%, well within the calculated measurement uncertainty. However, small but consistent discrepancies are visible at certain knob settings away from resonance: the descending sweep tends to produce marginally larger amplitudes at settings below resonance, such as knob settings 36 and 38, while the ascending sweep yields slightly larger values at settings above resonance, such as knob settings 52 and 54. These minor asymmetries between the two sweep directions are consistent with a small degree of nonlinear hysteresis in the system, likely attributable to the weakly nonlinear stiffening behavior of the spring and dry friction in the collar bearings, both of which are inherent to the physical apparatus. While the differences are largely within the bounds of experimental uncertainty, they are most pronounced in the vicinity of resonance where oscillation amplitudes are largest and nonlinear effects are most significant, which aligns with the expectations set out in the project manual regarding the weakly nonlinear character of the oscillator.

Figure 8 shows the theoretical phase lag between the upper mass displacement and the driving input as a function of driving frequency, computed using the experimentally determined values of $\omega_n = 15.916$ rad/s and $\zeta = 0.107$. The phase transitions smoothly from near zero at low frequencies to -180° at high frequencies, crossing -90° at the natural frequency, a behavior that is consistent with second-order linear oscillator theory and is marked by the dashed reference lines on the plot. The range of the plot is the range of knob frequencies found.

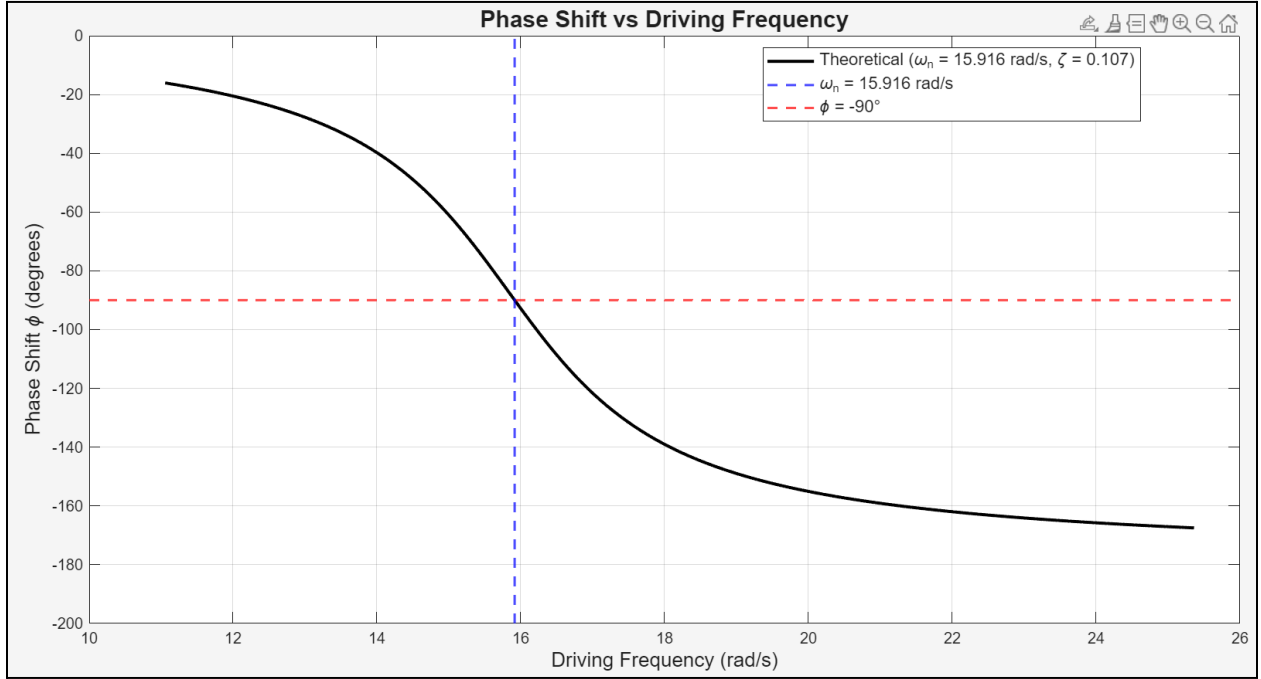


Figure 8. Theoretical phase shift as a function of driving frequency, computed using $\omega_n = 15.916$ rad/s and $\zeta = 0.107$; the blue dashed line marks the natural frequency, where the phase crosses -90° , and the red dashed line indicates the -90° reference.

Figure 9 presents the central result of this experiment: the experimentally measured magnitude ratio $M = \frac{X}{X_b}$ as a function of driving frequency for both the ascending and descending sweeps, plotted alongside the theoretical curve computed using $\omega_n = 15.916$ rad/s and $\zeta = 0.107$. The error bars reflect the combined systematic and random uncertainty calculated in the preceding section, and the close agreement between the experimental data and the theoretical curve confirms that the linear second-order model provides an accurate description of the system's frequency response over the range of frequencies tested.

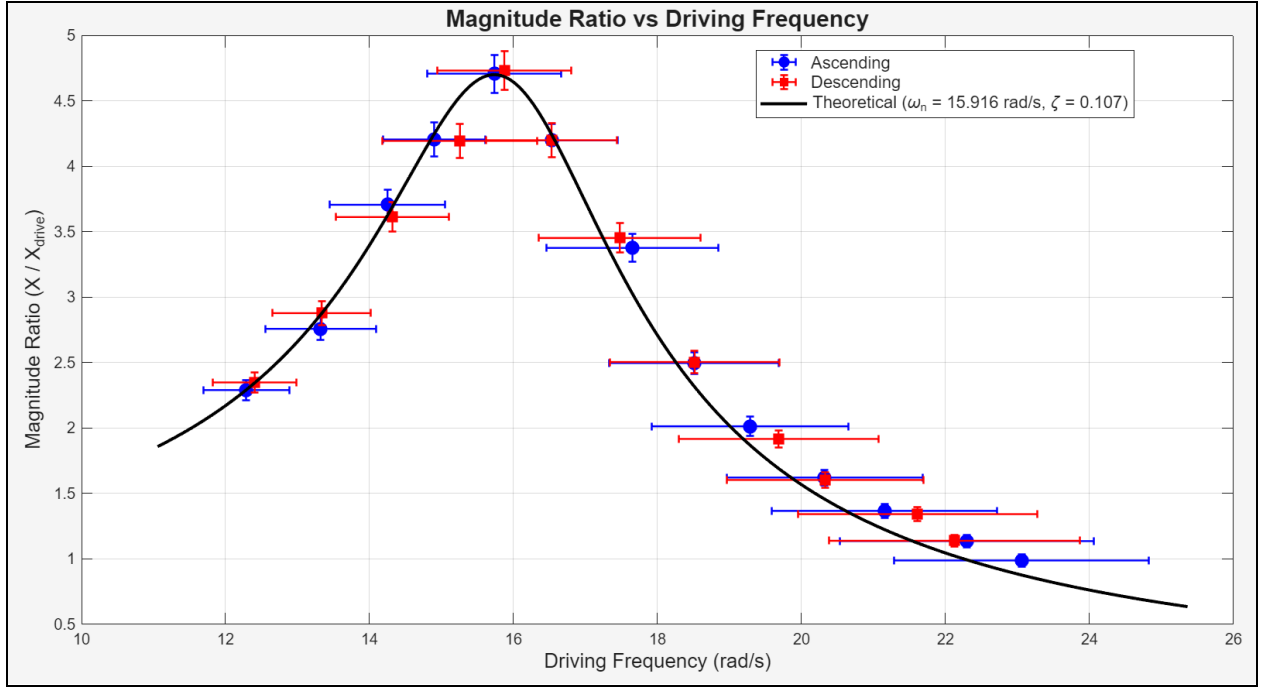


Figure 9. Experimental magnitude ratio ($\frac{X}{X_b}$) as a function of driving frequency for ascending (blue circles) and descending (red squares) frequency sweeps, plotted alongside the theoretical magnitude ratio curve computed using $\omega_n = 15.916$ rad/s and $\zeta = 0.107$.

Discussion

The experimental results obtained from the second-order spring-mass-damper system demonstrated behavior that aligned closely with the theoretical predictions described by the frequency response function of a second-order linear oscillator. As the driving frequency was incrementally increased from below the system's natural frequency toward resonance, a clear and progressive increase in the amplitude of the upper mass oscillation was observed. This trend was consistent with the theoretical amplitude ratio M as a function of frequency ratio ω/ω_n , which predicts that as the driving frequency approaches the natural frequency of the system, the denominator of the frequency response function approaches a minimum, causing the amplitude ratio to rise sharply toward its peak value.

As the driving frequency continued to increase and passed through the resonant frequency ω_R , the oscillation amplitude reached a well-defined maximum before beginning to decrease. This peak in amplitude response is the hallmark characteristic of resonance in a second-order system, and its experimental observation confirms that the system was being driven at or very near its natural frequency at that point in the sweep. The sharpness of this peak is directly related to the damping ratio ζ of the system. A lightly damped system produces a tall, narrow resonance peak, while a heavily damped

system produces a broader, lower peak. The shape of the observed peak can therefore be used in conjunction with the theoretical frequency response curves to estimate the system's damping ratio, serving as one of the two independent methods of damping ratio determination required for this experiment.

The damping ratio was determined via two independent methods, yielding values of $\zeta = 0.154$ from the logarithmic decrement and $\zeta = 0.107$ from the peak magnitude method. Method 2 is the preferred value due to its lower calculated uncertainty (± 0.003 versus ± 0.0125) and because Method 1 relies on only two peaks during a large-amplitude free decay where spring nonlinearity violates the constant-k assumption underlying the logarithmic decrement formula. The lower damping ratio obtained from Method 2 is physically consistent with this reasoning: at steady-state resonance the mass moves rapidly, meaning dry friction contributes proportionally less to total damping than during the slow free decay captured in Method 1. Additionally, spring nonlinearity at large oscillation amplitudes inflates M_{peak} beyond its true linear value, causing the damping ratio back-calculated from the peak magnitude formula to read slightly low. Method 1, by contrast, captures all damping sources during free decay and therefore reads higher.

The 29% discrepancy between the natural frequency determined via Method 1 ($\omega_n = 22.495 \text{ rad/s}$) and Method 2 ($\omega_n = 15.916 \text{ rad/s}$) is attributable to Method 1's reliance on the average static spring constant, which is inflated by nonlinear spring behavior at higher loads. The most physically relevant k value for the frequency sweep corresponds to the lighter loading regime (40.88–53.03 N/m), as the dynamic oscillation amplitudes during the sweep fall well below those imposed by the heavier static loads. The average k of 57.99 N/m therefore overestimates the effective stiffness during operation, causing Method 1 to systematically overpredict ω_n . Method 2, derived directly from the observed resonance frequency and damping ratio, better reflects the system's true dynamic behavior and is therefore used as the basis for the theoretical frequency response curve. A third independent estimate of $\omega_n = 16.526 \text{ rad/s}$, back-calculated from the step response damped frequency, agrees with Method 2 to within 1.1%, corroborating this choice.

The experimental results demonstrate strong internal self-consistency across all three methods of natural frequency determination. The natural frequency determined from the step response, $f_n = 2.630 \text{ Hz}$ ($\omega_n = 16.526 \text{ rad/s}$), closely matches the driving frequency at the resonance knob setting of $f_r = 2.599 \text{ Hz}$, a difference of only 1.2%, confirming that the system was indeed driven at resonance at that knob setting. Furthermore, the resonance frequency $\omega_r = 16.336 \text{ rad/s}$ and natural frequency $\omega_n = 16.526 \text{ rad/s}$ differ by only 1.1%, which is precisely what is expected for a

lightly damped system. The theoretical relationship $w_r = w_n \sqrt{1 - 2\zeta^2}$ predicts these two quantities to be nearly identical when ζ is small. Together, these agreements validate the internal coherence of the measured system parameters.

Once the driving frequency was increased beyond resonance, the oscillation amplitude of the upper mass declined steadily, even as the motor continued to spin faster. This post-resonance attenuation is also consistent with linear second-order theory. At frequencies significantly above ω_n , the inertia of the mass prevents it from responding to the rapidly oscillating input, and the amplitude ratio M asymptotically approaches zero. In physical terms, the system simply cannot "keep up" with the driving force at high frequencies, and the upper mass effectively becomes decoupled from the base excitation. This behavior was clearly visible in the data as the amplitude trended downward through the upper frequency range of the sweep.

It is also worth noting that data was collected for both increasing and decreasing frequency sweeps. In a perfectly linear system, the amplitude response should be identical regardless of the direction of the frequency sweep. Any observed discrepancy between the two sweep directions could indicate the presence of non-linear behavior in the system, such as geometric non-linearity in the spring or dry friction in the collar bearings, both of which are inherent to a real physical apparatus of this type. The amplitude values at resonance were 0.2745 m and 0.2759 m for the ascending and descending sweeps respectively, a difference of less than 0.6%, well within the calculated measurement uncertainty. However, small but consistent discrepancies are visible at certain knob settings away from resonance: the descending sweep tends to produce marginally larger amplitudes at settings below resonance, such as knob settings 36 and 38, while the ascending sweep yields slightly larger values at settings above resonance, such as knob settings 52 and 54. These minor asymmetries between the two sweep directions are consistent with a small degree of nonlinear hysteresis in the system, likely attributable to the weakly nonlinear stiffening behavior of the spring and dry friction in the collar bearings, both of which are inherent to the physical apparatus. While the differences are largely within the bounds of experimental uncertainty, they are most pronounced in the vicinity of resonance where oscillation amplitudes are largest and nonlinear effects are most significant.

Conclusion

The primary goal of this experiment was to utilize an ultrasonic air-ranging sensor to accurately characterize the dynamic behavior of a second-order spring-mass-damper system through experimental measurement of its frequency response. Based on the results obtained and the observations made throughout the data acquisition process, this goal was

successfully achieved. The ultrasonic sensor proved to be a capable and practical transducer for this application, providing non-contact, direct displacement measurements of the oscillating upper mass without imposing any mechanical loading on the system itself, a significant advantage over contact-based sensors such as the linear potentiometer.

The spring constant of the system was determined via static deflection, and the natural frequency and damping ratio were each established through two independent methods, providing a degree of internal consistency and cross-validation of the system parameters. These experimentally determined values formed the foundation of the theoretical frequency response curve against which the measured data was compared. The system's response across the full range of driving frequencies was captured with sufficient frequency resolution to produce a well-defined response curve, consistent with the behavior predicted by linear second-order oscillator theory.

The most significant and visually apparent result of the experiment was the clear amplitude peak observed as the driving frequency approached the system's resonant frequency, followed by a steady decline in amplitude as the frequency was swept beyond resonance. This behavior matched the theoretical predictions described by the amplitude ratio M as a function of the frequency ratio ω/ω_n , validating the linear second-order model as a reasonable representation of the physical system over the range of frequencies tested. The damping ratio estimated from the shape of the resonance peak was consistent with the values obtained through the other determination methods, further supporting the internal coherence of the results.

Some degree of deviation from ideal linear behavior was expected and observed, arising from inherent non-linearities in the physical apparatus such as bearing friction, spring non-linearity, and small geometric imperfections in the setup. These deviations were most noticeable in the vicinity of resonance, where oscillation amplitudes were largest and the system was most sensitive to non-linear effects. While the linear model does not capture these phenomena explicitly, it nonetheless provided an accurate and useful first-order description of the system's dynamic behavior across the majority of the frequency range tested.

From an instrumentation standpoint, this experiment highlighted both the strengths and the practical considerations associated with ultrasonic ranging as a measurement technique. The sensor's non-contact nature, direct displacement output, and freedom from calibration made it well suited for this application. However, careful attention to beam alignment, sampling rate selection relative to the sensor's repetition frequency, and the temperature dependence of the speed of sound were all identified as important factors in obtaining high quality data. Overall, the ultrasonic sensor successfully fulfilled its role as the primary measurement instrument, and the experiment demonstrated that with proper

setup and data acquisition practices, a second-order mechanical system can be accurately modeled and its key dynamic parameters reliably extracted from experimental frequency response data.

References

ME310 Lab Manual – Project: Design and Implementation of a Transduction System for the Measurement of a Weakly Nonlinear Mechanical Oscillator
by Drs. B. Bharadvaj, M. Isaacson, R.G. Holt, and C.H. Farny, Jan. 2026.

Figliola, Richard S., and Donald E. Beasley. *Theory and Design for Mechanical Measurements*. Fifth Edition, John Wiley and Sons, inc., 2011.

KeySight “34661A Digital Multimeter Data Sheet,” Nov 2025,
<https://www.keysight.com/us/en/assets/7018-03846/data-sheets/5991-1983.pdf>

National Instruments “BNC 2090A User Manual,” Jan 2007,
<https://www.ni.com/docs/en-US/bundle/bnc-2090a-feature/resource/372101a.pdf>

KeySight “3631A Power Supply Data Sheet,” May 2018,
<https://assets-us-01.kc-usercontent.com/ecb176a6-5a2e-0000-8943-84491e5fc8d1/8a49e6d7-769b-4a6b-a093-b685fa1c106f/Keysight%20E363xA%20Series%20Programmable%20DC%20Power%20Supplies%20Data%20Sheet.pdf>

Weight Set
[https://us.ohaus.com/en-us/products/weights/reference-weights-and-mass-standards/astm-reference-calibration-weight-sets/weight-set\(21\)-100-g-1-mg-ultraclass](https://us.ohaus.com/en-us/products/weights/reference-weights-and-mass-standards/astm-reference-calibration-weight-sets/weight-set(21)-100-g-1-mg-ultraclass)

Power
<https://hypertextbook.com/facts/2000/CheukWong.shtml#:~:text=The%20lower%20the%20density%20that,Cheuk%20Wong%20%2D%2D%202000>